

Status note for arXiv:1602.00473, Question 3.11

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Status

This is a lightweight literature-status packet. The source question is already answered in a later paper by Candeloro–Di Piazza–Musial–Sambucini; no new proof is claimed here.

Source question

The source paper is D. Candeloro, L. Di Piazza, K. Musial, and A. R. Sambucini, *Gauge integrals and selections of weakly compact valued multifunctions*, J. Math. Anal. Appl. 441 (1) (2016), 293–308, DOI 10.1016/j.jmaa.2016.04.009, arXiv:1602.00473.

In Section 3, Question 3.11 asks whether there exists at least one variationally Henstock integrable selection of a $cwk(X)$ -valued variationally Henstock integrable multifunction. Immediately after the question the authors say that the answer was not known for an arbitrary Banach space, while it was known under the Radon–Nikodym property.

Answering theorem

The answer source is D. Candeloro, L. Di Piazza, K. Musial, and A. R. Sambucini, *Relations among Gauge and Pettis integrals for $cwk(X)$ -valued multifunctions*, Annali di Matematica Pura ed Applicata 197 (1) (2018), 171–183, DOI 10.1007/s10231-017-0674-z, available locally as arXiv:1912.01845.

The supporting paper’s introduction states that its Theorem 4.1 solves the existence problem for variationally Henstock integrable selections of $cwk(X)$ -valued variationally Henstock integrable multifunctions, citing the source as Question 3.11. In Section 4 it then says it examines Question 3.11, removes the Radon–Nikodym-property hypothesis from the earlier partial theorem, and gives a complete answer.

Theorem 4.1 states: if X is any Banach space and $\Gamma : [0, 1] \rightarrow cwk(X)$ is variationally Henstock integrable, then $\mathcal{S}_{vH}(\Gamma) \neq \emptyset$, and every strongly measurable selection of Γ is variationally Henstock integrable. Therefore Question 3.11 has an affirmative answer in arbitrary Banach spaces.

Provenance check

Crossref metadata for the supporting paper records the source paper as a reference by DOI 10.1016/j.jmaa.2016.04.009. The supporting paper also names the source question explicitly, so this is an exact later answer rather than an agent-identified reformulation.

Conclusion

The target arXiv:1602.00473 is not viable for a new proof or counterexample on Question 3.11. The question is already answered affirmatively by Theorem 4.1 of arXiv:1912.01845 / Annali di Matematica Pura ed Applicata 197 (2018).

References

- [1] D. Candeloro, L. Di Piazza, K. Musiał, and A. R. Sambucini, *Gauge integrals and selections of weakly compact valued multifunctions*, J. Math. Anal. Appl. 441 (1) (2016), 293–308.
- [2] D. Candeloro, L. Di Piazza, K. Musiał, and A. R. Sambucini, *Relations among Gauge and Pettis integrals for $cwk(X)$ -valued multifunctions*, Annali di Matematica Pura ed Applicata 197 (1) (2018), 171–183.